Lab 1: Dockerize and Serve a Pretrained Model

1. Pretrained Model Selection and Serving

For this lab I selected the DistilBERT pretrained model (distilbert-base-uncased-finetuned-sst-2-english) from the Hugging Face library. This model is fine-tuned for text classification specifically sentiment analysis.

To serve the model I built a REST API using the Flask web framework in Python. The application defines a single POST endpoint (/predict) that accepts a JSON payload containing text processes it through the Hugging Face pipeline and returns the sentiment prediction and confidence score. To ensure the model is served using a production-quality server framework as required by the rubric the Flask app is wrapped and served using Waitress.

2. Dockerization

The application was containerized using a Dockerfile based on the official python:3.9-slim lightweight image. The Dockerfile installs the necessary dependencies (Flask Waitress PyTorch Transformers) via requirements.txt copies the application code into the container and exposes port 5000.

The image was successfully built and pushed to Docker Hub.

Docker Hub Link: https://hub.docker.com/r/direraven22740/model-service

3. Visual Verification

A. Running the Container Locally

The screenshot below demonstrates the Docker container successfully running the Waitress server on port 5000 locally.

```
PS D:\SEG4180 Applied Machine Learning Assignments And Project\seg4180-lab1> docker run -p 5000:5000 DireRaven22740/model-service:latest
Device set to use cpu
Loading DistilBERT model... This might take a moment on the first run.
Starting server on port 5000...
```

B. Sample API Request and Prediction

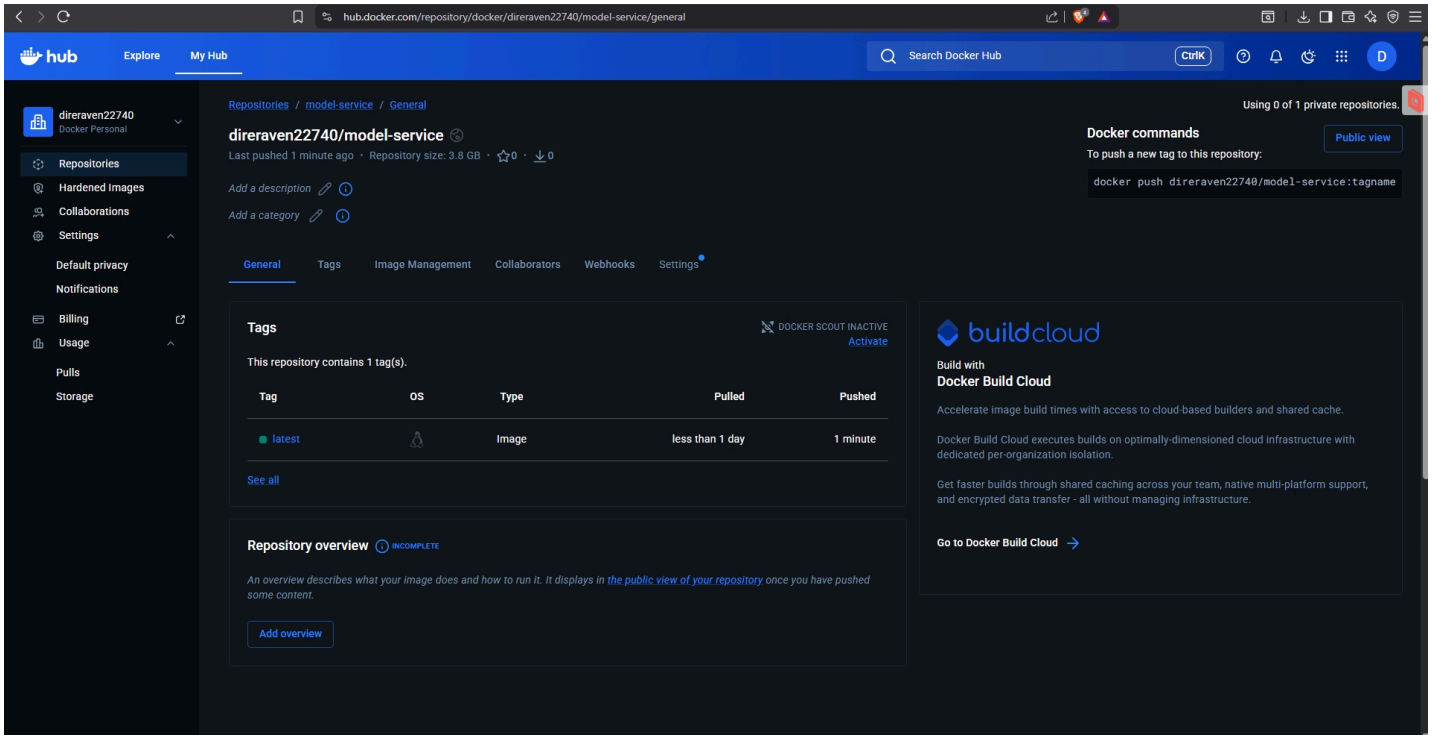
The screenshot below demonstrates a local POST request sent to the containerized API using PowerShell's Invoke-RestMethod. The model successfully received the input and returned a JSON response predicting a "POSITIVE" sentiment with 99.98% confidence.

```
PS D:\SEG4180 Applied Machine Learning Assignments And Project\seg4180-lab1> Invoke-RestMethod -Uri http://localhost:5000/predict -Method Post -Body '{"text": "Playing games is an unforgettable experience."}' -ContentType "application/json"

        confidence input                                  prediction
        ---------- -----                                  ----------
0.9998664855957031 Playing games is an unforgettable experience. POSITIVE
```

C. Docker Hub Repository

The screenshot below verifies the Docker image was successfully tagged and pushed to the public Docker Hub registry.



4. Instructions to Run the Container Locally
To pull and run this model on any machine execute the following steps:

Ensure Docker is running.

Open a terminal and pull the image: docker pull direraven22740/model-service:latest

Run the container and map the ports: docker run -p 5000:5000 direraven22740/model-service:latest

In a new terminal test the endpoint by sending a POST request to http://localhost:5000/predict with a JSON payload containing a "text" key.